Mini Project 2 Big Data Report

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1 Problem Statement

The aviation industry generates enormous volumes of operational data daily, yet flight delays remain one of the most persistent and costly inefficiencies in air travel. In 2019–2023, tens of millions of US domestic flights were affected by delays attributable to a range of causes including carrier operations, weather events, National Airspace System congestion, security incidents, and late-arriving aircraft.

Research Question: *What drives flight delay severity across US airlines and routes, and how do delay causes propagate over time?*

Specifically, we investigate three analytical dimensions:

- **Trend Analysis** – Temporal delay patterns using 7-day moving averages and cumulative cancellation curves to identify seasonal spikes and long-term airline degradation trends.
- **Anomaly Detection** – Statistical identification of route-level delay outliers (departures exceeding $\mu + 3\sigma$ of the route mean), enabling targeted infrastructure or scheduling interventions.
- **Root Cause Attribution** – Decomposing aggregate delay minutes by cause (carrier, weather, NAS, security, late aircraft) per airline to prioritise improvement efforts.

The dataset's 3 million records and 32 mixed-type columns provide the scale and richness needed to stress-test Apache Spark's distributed execution model, compare its three programming APIs (SQL, DataFrame, RDD), and demonstrate the concrete performance gains delivered by the Catalyst Optimizer and Tungsten execution engine.

2 Dataset Description

We used US Flight Delay and Cancellation Data from Kaggle, it contains 3M records, and it's schema is defined as in the following table:

Column Name	Data Type
FL_DATE	Date
AIRLINE	String
AIRLINE_DOT	String
AIRLINE_CODE	String
DOT_CODE	Integer
FL_NUMBER	Integer
ORIGIN	String
ORIGIN_CITY	String
DEST	String
DEST_CITY	String
CRS_DEP_TIME	Integer
DEP_TIME	Double
DEP_DELAY	Double
TAXI_OUT	Double
WHEELS_OFF	Double
WHEELS_ON	Double
TAXI_IN	Double
CRS_ARR_TIME	Integer
ARR_TIME	Double
ARR_DELAY	Double
CANCELLED	Double
CANCELLATION_CODE	String
DIVERTED	Double
CRS_ELAPSED_TIME	Double
ELAPSED_TIME	Double
AIR_TIME	Double
DISTANCE	Double
DELAY_DUE_CARRIER	Double
DELAY_DUE_WEATHER	Double
DELAY_DUE_NAS	Double
DELAY_DUE_SECURITY	Double
DELAY_DUE_LATE_AIRCRAFT	Double

Table 1: Schema of Flights Dataset

2.1 Justification

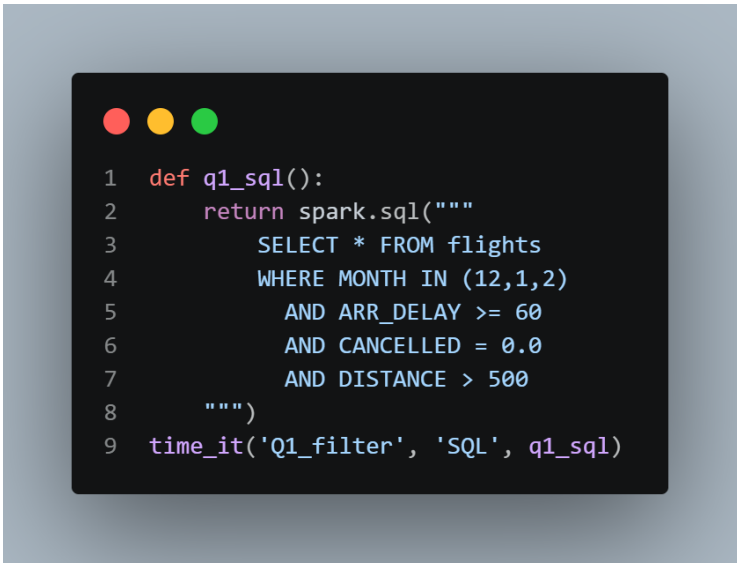
This data is appropriate to be distributed processed due to the presence of 3 million records, which is efficient to be processed with the help of such frameworks as Spark to carry out complex operations (joins, aggregations, window functions, etc.). Moreover, it contains 32 mixed-columns with both numerical, categorical and date values, which can be helpful in testing Spark SQL and query optimization abilities. It also inherently supports join operations to dimension tables, like airlines and airports, and time-related capabilities allow more complex analytics, like window functions, moving averages, and cumula-

tive computations.

3 Queries Implemented

3.1 Q1

3.1.1 SQL



```
def q1_sql():
    return spark.sql("""
        SELECT * FROM flights
        WHERE MONTH IN (12,1,2)
          AND ARR_DELAY >= 60
          AND CANCELLED = 0.0
          AND DISTANCE > 500
    """)
time_it('Q1_filter', 'SQL', q1_sql)
```

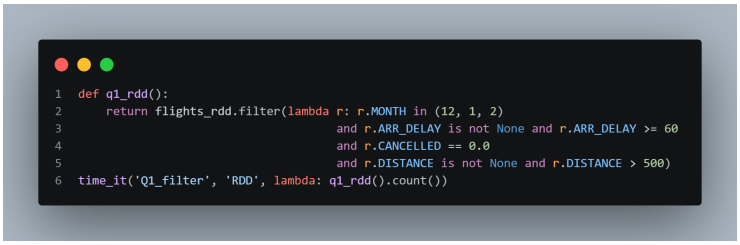
Figure 1: Q1 - SQL Implementation

3.1.2 Dataframe

```
def q1_df():
    return (flights
            .filter(F.col('MONTH').isin(12, 1, 2))
            .filter(F.col('ARR_DELAY') >= 60)
            .filter(F.col('CANCELLED') == 0.0)
            .filter(F.col('DISTANCE') > 500))
q1_df_res = time_it('Q1_filter', 'DataFrame', q1_df)

```

Figure 2: Q1 - DataFrame Implementation

3.1.3 RDD



```
def q1_rdd():
    return flights_rdd.filter(lambda r: r.MONTH in (12, 1, 2)
                                        and r.ARR_DELAY is not None and r.ARR_DELAY >= 60
                                        and r.CANCELLED == 0.0
                                        and r.DISTANCE is not None and r.DISTANCE > 500)
time_it('Q1_filter', 'RDD', lambda: q1_rdd().count())
```

Figure 3: Q1 - RDD Implementation

3.1.4 Q1 Explain

The **parsed logical plan** expresses Q1 as a chain of four `Filter` nodes (long-haul, severe arrival delay, winter month, non-cancelled) above a CSV `Relation`. The **optimized logical plan** collapses the four predicates into a single conjunction and adds `isnotnull` guards. The **physical plan** is a single-stage `FileScan csv` with a `PushedFilters` clause carrying every predicate, followed by an in-memory `Project` of three columns – no `Exchange`, no shuffle.

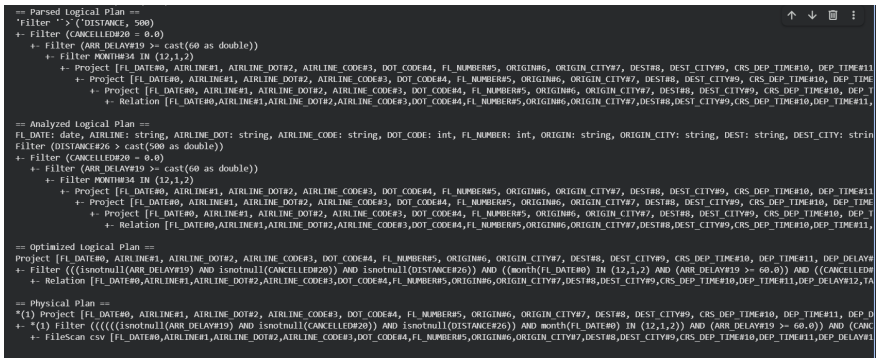


Figure 4: Q1 Explain

3.2 Q2

3.2.1 SQL

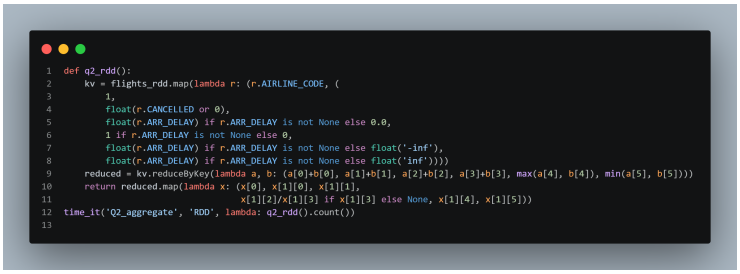
Figure 5: Q2 - SQL Implementation

3.2.2 Dataframe

```python
def q2_df():
    return (flights.groupBy('AIRLINE_CODE')
            .agg(F.count('*').alias('total_flights'),
                 F.sum('CANCELLED').alias('cancellations'),
                 F.avg('ARR_DELAY').alias('avg_arr_delay'),
                 F.max('ARR_DELAY').alias('max_arr_delay'),
                 F.min('ARR_DELAY').alias('min_arr_delay')))
q2_df_res = time_it('Q2_aggregate', 'DataFrame', q2_df)

```

Figure 6: Q2 - DataFrame Implementation

3.2.3 RDD



```python
def q2_rdd():
    kv = flights_rdd.map(lambda r: (r.AIRLINE_CODE, (
        1,
        float(r.CANCELLED or 0),
        float(r.ARR_DELAY) if r.ARR_DELAY is not None else 0.0,
        1 if r.ARR_DELAY is not None else 0,
        float(r.ARR_DELAY) if r.ARR_DELAY is not None else float('-inf'),
        float(r.ARR_DELAY) if r.ARR_DELAY is not None else float('inf'))))
    reduced = kv.reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1], a[2]+b[2], a[3]+b[3], max(a[4], b[4]), min(a[5], b[5])))
    return reduced.map(lambda x: (x[0], x[1][0], x[1][1],
                                  x[1][2]/x[1][3] if x[1][3] else None, x[1][4], x[1][5]))
time_it('Q2_aggregate', 'RDD', lambda: q2_rdd().count())

```

Figure 7: Q2 - RDD Implementation

3.2.4 Q2 Explain

The **parsed logical plan** groups the full row-set by `AIRLINE_CODE` and computes `avg(ARR_DELAY)`. The **optimized logical plan** prunes the schema down to the two needed columns and pushes an `isnotnull(ARR_DELAY)` filter into the file scan. The **physical plan** adopts the canonical two-stage `HashAggregate`: a partial aggregation per partition, an `Exchange hashpartitioning(AIRLINE_CODE, 200)` shuffle, then a final `HashAggregate` to merge the partials.

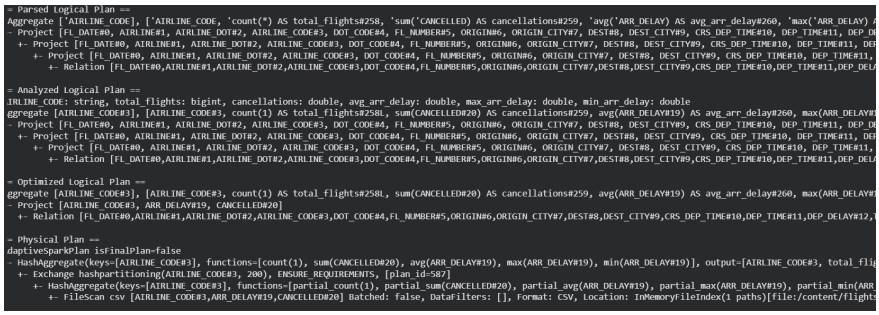


Figure 8: Q2 Explain

3.3 Q3

3.3.1 SQL

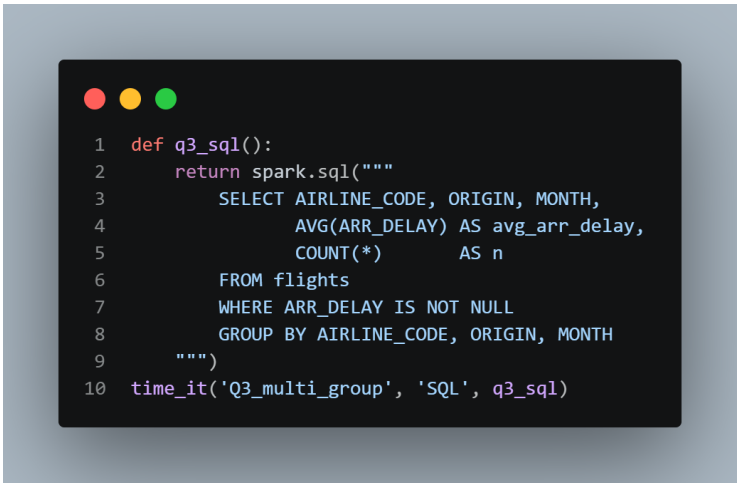
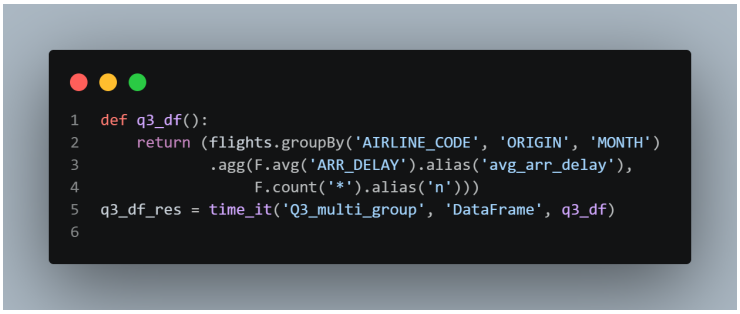


Figure 9: Q3 - SQL Implementation

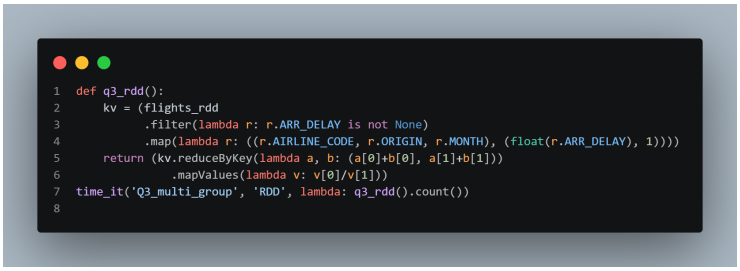
3.3.2 Dataframe



```
def q3_df():
    return (flights.groupBy('AIRLINE_CODE', 'ORIGIN', 'MONTH')
            .agg(F.avg('ARR_DELAY').alias('avg_arr_delay'),
                 F.count('*').alias('n')))
q3_df_res = time_it('Q3_multi_group', 'DataFrame', q3_df)

```

Figure 10: Q3 - DataFrame Implementation

3.3.3 RDD



```
def q3_rdd():
    kv = (flights_rdd
          .filter(lambda r: r.ARR_DELAY is not None)
          .map(lambda r: ((r.AIRLINE_CODE, r.ORIGIN, r.MONTH), (float(r.ARR_DELAY), 1))))
    return (kv.reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
              .mapValues(lambda v: v[0]/v[1]))
time_it('Q3_multi_group', 'RDD', lambda: q3_rdd().count())

```

Figure 11: Q3 - RDD Implementation

3.3.4 Q3 Explain

The **parsed logical plan** derives a `MONTH` column then groups by three keys (`AIRLINE_CODE`, `ORIGIN`, `MONTH`). The **optimized logical plan** pushes the `MONTH` computation below the aggregation, prunes columns to the four needed, and inserts `isnotnull` filters at scan. The **physical plan** exhibits two-stage `HashAggregate` with a wide `Exchange hashpartitioning(AIRLINE_CODE, ORIGIN, MONTH, 200)` – the high-cardinality composite key makes shuffle the dominant cost.

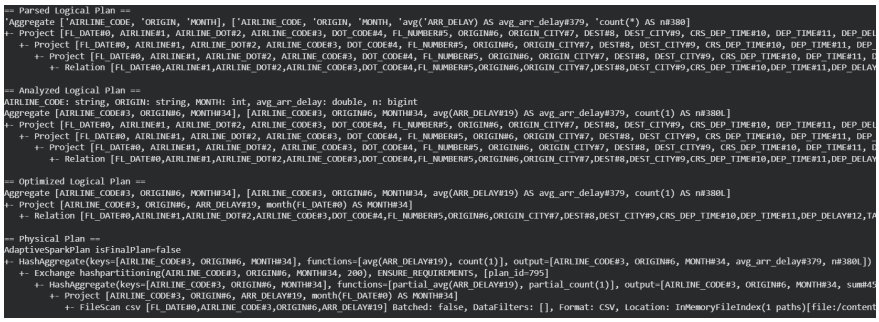


Figure 12: Q3 Explain

3.4 Q4

3.4.1 SQL

```
def q4_sql():
    return spark.sql("""
        SELECT ORIGIN, DEST,
               AVG(ARR_DELAY) AS avg_delay,
               COUNT(*)       AS flights
        FROM flights
        GROUP BY ORIGIN, DEST
        HAVING COUNT(*) > 100
        ORDER BY avg_delay DESC
        LIMIT 20
    """)
time_it('Q4_top20_routes', 'SQL', q4_sql)

```

Figure 13: Q4 - SQL Implementation

3.4.2 Dataframe

```python
def q4_df():
    return (flights.groupBy('ORIGIN', 'DEST')
            .agg(F.avg('ARR_DELAY').alias('avg_delay'),
                 F.count('*').alias('flights'))
            .filter('flights > 100')
            .orderBy(F.desc('avg_delay'))
            .limit(20))
q4_df_res = time_it('Q4_top20_routes', 'DataFrame', q4_df)

```

Figure 14: Q4 - DataFrame Implementation

3.4.3 RDD

```python
def q4_rdd():
    kv = (flights_rdd
          .filter(lambda r: r.ARR_DELAY is not None)
          .map(lambda r: ((r.ORIGIN, r.DEST), (float(r.ARR_DELAY), 1))))
    agg = kv.reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1])).mapValues(lambda v: v[0]/v[1])
    return agg.takeOrdered(20, key=lambda x: -x[1])
time_it('Q4_top20_routes', 'RDD', lambda: len(q4_rdd()))

```

Figure 15: Q4 - RDD Implementation

3.4.4 Q4 Explain

The **parsed logical plan** groups by (`ORIGIN`, `DEST`), filters `flights > 100`, sorts by average delay descending, and limits to 20. The **optimized logical plan** prunes to three columns and reorders the post-aggregation `Filter` after the `HashAggregate` (it cannot be pushed below the group-by). The **physical plan** replaces the naive `Sort + Limit` with a single `TakeOrderedAndProject(20)` operator that performs a partial top-20 per partition before a final merge – avoiding a full shuffle-sort on 24k aggregated rows.

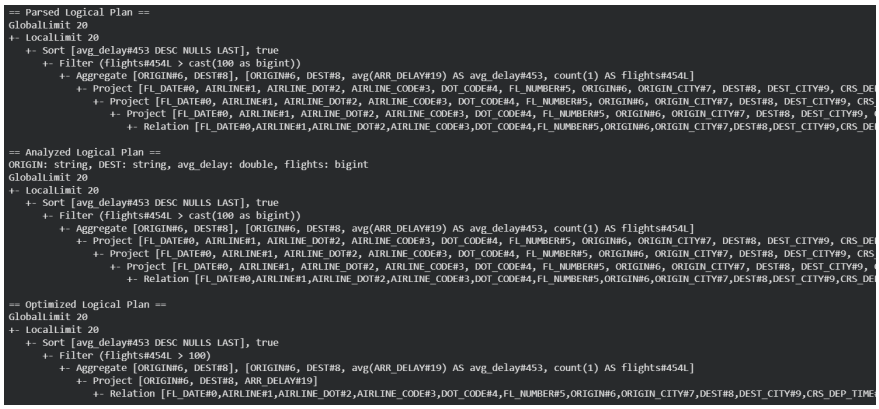


Figure 16: Q4 Explain

3.5 Q5

3.5.1 SQL

```sql
def q5_sql():
    return spark.sql("""
        WITH daily AS (
          SELECT AIRLINE_CODE, FL_DATE, AVG(ARR_DELAY) AS daily_avg
          FROM flights WHERE ARR_DELAY IS NOT NULL
          GROUP BY AIRLINE_CODE, FL_DATE
        )
        SELECT AIRLINE_CODE, FL_DATE, daily_avg,
               AVG(daily_avg) OVER (PARTITION BY AIRLINE_CODE
                                    ORDER BY FL_DATE
                                    ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) AS ma7
        FROM daily
    """)
time_it('Q5_moving_avg', 'SQL', q5_sql)

```

Figure 17: Q5 - SQL Implementation

3.5.2 Dataframe

```python
def q5_df():
    daily = (flights.filter(F.col('ARR_DELAY').isNotNull())
             .groupBy('AIRLINE_CODE', 'FL_DATE')
             .agg(F.avg('ARR_DELAY').alias('daily_avg')))
    w = Window.partitionBy('AIRLINE_CODE').orderBy('FL_DATE').rowsBetween(-6, 0)
    return daily.withColumn('ma7', F.avg('daily_avg').over(w))
q5_df_res = time_it('Q5_moving_avg', 'DataFrame', q5_df)

```

Figure 18: Q5 - DataFrame Implementation

3.5.3 RDD

```python
def q5_rdd():
    daily = (flights_rdd
             .filter(lambda r: r.ARR_DELAY is not None)
             .map(lambda r: ((r.AIRLINE_CODE, r.FL_DATE), (float(r.ARR_DELAY), 1)))
             .reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
             .map(lambda x: (x[0][0], (x[0][1], x[1][0]/x[1][1]))))
    grouped = daily.groupByKey().mapValues(lambda it: sorted(it))
    def rolling(pairs):
        out = []
        for i in range(len(pairs)):
            window = pairs[max(0, i-6):i+1]
            out.append((pairs[i][0], sum(p[1] for p in window)/len(window)))
        return out
    return grouped.flatMapValues(rolling)
time_it('Q5_moving_avg', 'RDD', lambda: q5_rdd().count())

```

Figure 19: Q5 - RDD Implementation

3.5.4 Q5 Explain

The **parsed logical plan** contains a `groupBy(AIRLINE_CODE, FL_DATE)` averaging `ARR_DELAY` feeding a `Window` with `ROWS BETWEEN 6 PRECEDING AND CURRENT ROW`. The **optimized logical plan** prunes to three columns and pushes the `isnotnull(ARR_DELAY)` guard to the file scan. The **physical plan** chains two `Exchange` stages: first `hashpartitioning(AIRLINE_CODE, FL_DATE, 200)` for the aggregation, then a re-partition to `hashpartitioning(AIRLINE_CODE, 200)` followed by a `Sort(FL_DATE)` so the `Window` operator can sweep each partition in order.

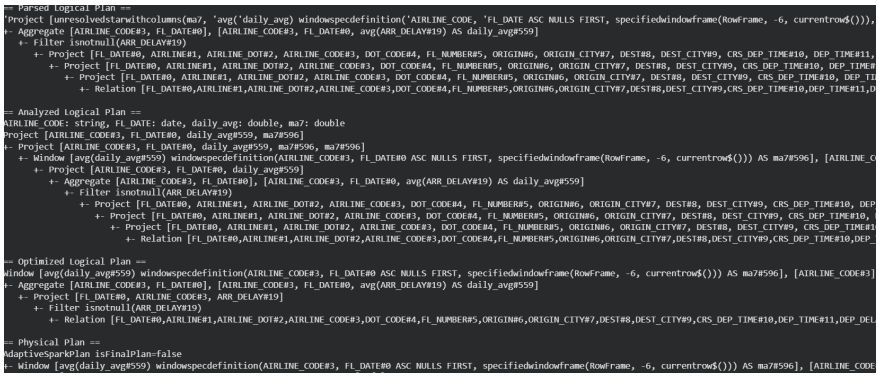


Figure 20: Q5 Explain

3.6 Q6

3.6.1 SQL

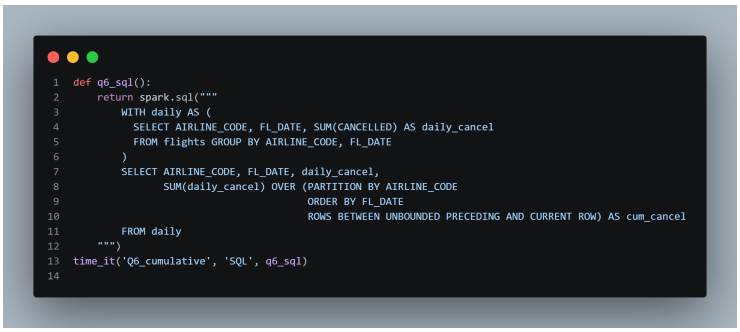


Figure 21: Q6 - SQL Implementation

3.6.2 Dataframe

```python
def q6_df():
    daily = (flights.groupBy('AIRLINE_CODE', 'FL_DATE')
             .agg(F.sum('CANCELLED').alias('daily_cancel')))
    w = (Window.partitionBy('AIRLINE_CODE').orderBy('FL_DATE')
         .rowsBetween(Window.unboundedPreceding, Window.currentRow))
    return daily.withColumn('cum_cancel', F.sum('daily_cancel').over(w))
q6_df_res = time_it('Q6_cumulative', 'DataFrame', q6_df)

```

Figure 22: Q6 - DataFrame Implementation

3.6.3 RDD

```python
def q6_rdd():
    daily = (flights_rdd
             .map(lambda r: ((r.AIRLINE_CODE, r.FL_DATE), float(r.CANCELLED or 0)))
             .reduceByKey(lambda a, b: a + b)
             .map(lambda x: (x[0][0], (x[0][1], x[1]))))
    grouped = daily.groupByKey().mapValues(lambda it: sorted(it))
    def cumul(pairs):
        total = 0.0; out = []
        for d, c in pairs:
            total += c; out.append((d, total))
        return out
    return grouped.flatMapValues(cumul)
time_it('Q6_cumulative', 'RDD', lambda: q6_rdd().count())

```

Figure 23: Q6 - RDD Implementation

3.6.4 Q6 Explain

The **parsed logical plan** sums `CANCELLED` per (`AIRLINE_CODE`, `FL_DATE`) then applies an unbounded preceding window for the cumulative count. The **optimized logical plan** prunes to those three columns and folds away unused projections. The **physical plan** mirrors Q5 – two `Exchange` stages plus a `Sort(FL_DATE)` – but the window operator uses a *growing* frame, so memory cost per partition increases linearly with rows; Catalyst still uses a single streaming pass rather than recomputing the prefix sum from scratch.

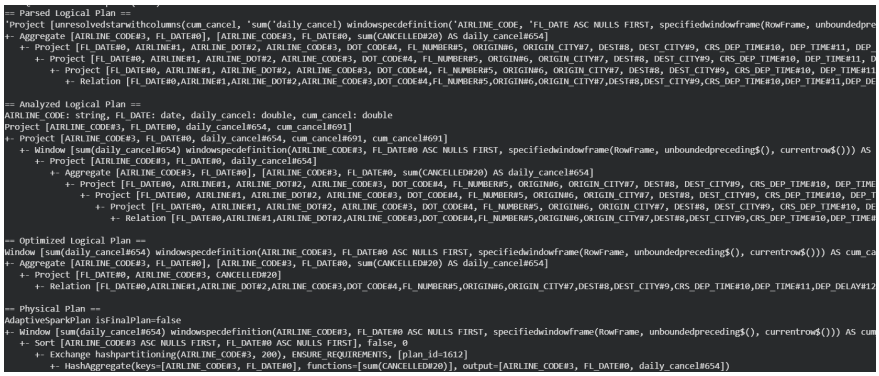


Figure 24: Q6 Explain

3.7 Q7

3.7.1 SQL

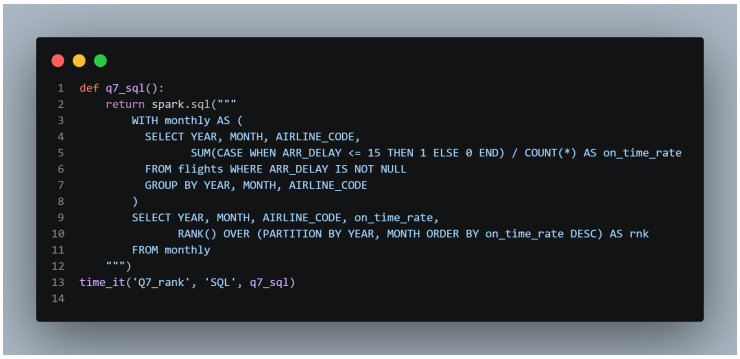


Figure 25: Q7 - SQL Implementation

3.7.2 Dataframe

Figure 26: Q7 - DataFrame Implementation

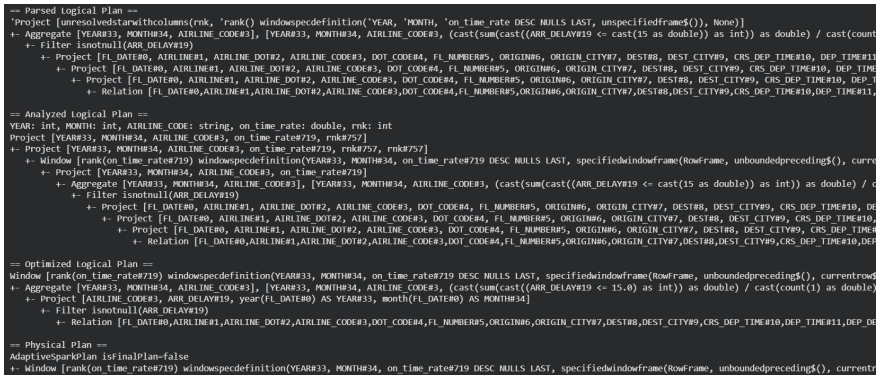
3.7.3 RDD

```python
def q7_rdd():
    kv = (flights_rdd
          .filter(lambda r: r.ARR_DELAY is not None)
          .map(lambda r: ((r.YEAR, r.MONTH, r.AIRLINE_CODE),
                          (1, 1 if r.ARR_DELAY <= 15 else 0))))
    agg = (kv.reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
             .map(lambda x: (x[0][0], x[0][1], x[0][2], x[1][1]/x[1][0])))
    grouped = agg.map(lambda x: ((x[0], x[1]), (x[2], x[3]))).groupByKey()
    def rank(rows):
        s = sorted(rows, key=lambda r: -r[1])
        return [(airline, rate, i+1) for i, (airline, rate) in enumerate(s)]
    return grouped.flatMapValues(rank)
time_it('Q7_rank', 'RDD', lambda: q7_rdd().count())
```

Figure 27: Q7 - RDD Implementation

3.7.4 Q7 Explain

The **parsed logical plan** derives `YEAR`/`MONTH`, computes the on-time rate as a ratio of conditional sums per (`YEAR`, `MONTH`, `AIRLINE_CODE`), then applies a `RANK()` window partitioned by (`YEAR`, `MONTH`). The **optimized logical plan** prunes to two source columns plus the two derived ones and pushes `isnotnull(ARR_DELAY)`. The **physical plan** runs a two-stage `HashAggregate` via `hashpartitioning(YEAR, MONTH, AIRLINE_CODE, 200)`, then re-shuffles to `hashpartitioning(YEAR, MONTH, 200)` and `Sort`s by descending on-time-rate before the streaming `RANK` window.

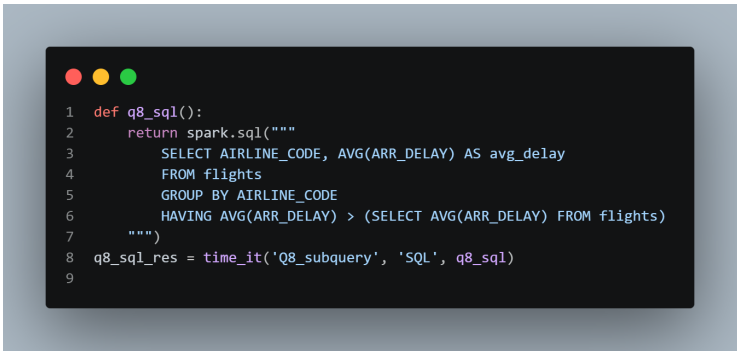


```
== Parsed Logical Plan ==
'Project [unresolvedstarwithcolumns(rnk, 'rank() windowspecdefinition('YEAR, 'MONTH, 'on_time_rate DESC NULLS LAST, unspecifiedframe$()), None)]
+- Aggregate [YEAR#33, MONTH#34, AIRLINE_CODE#3], [YEAR#33, MONTH#34, AIRLINE_CODE#3, (cast(sum(cast((ARR_DELAY#19 <= cast(15 as double)) as int)) as double) / cast(count
   +- Filter isnotnull(ARR_DELAY#19)
      +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME
            +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_T
               +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11,

== Analyzed Logical Plan ==
YEAR: int, MONTH: int, AIRLINE_CODE: string, on_time_rate: double, rnk: int
Project [YEAR#33, MONTH#34, AIRLINE_CODE#3, on_time_rate#719, rnk#757]
+- Project [YEAR#33, MONTH#34, AIRLINE_CODE#3, on_time_rate#719, rnk#757, rnk#757]
   +- Window [rank(on_time_rate#719) windowspecdefinition(YEAR#33, MONTH#34, on_time_rate#719 DESC NULLS LAST, specifiedwindowframe(RowFrame, unboundedpreceding$(), curre
      +- Project [YEAR#33, MONTH#34, AIRLINE_CODE#3, on_time_rate#719]
         +- Aggregate [YEAR#33, MONTH#34, AIRLINE_CODE#3], [YEAR#33, MONTH#34, AIRLINE_CODE#3, (cast(sum(cast((ARR_DELAY#19 <= cast(15 as double)) as int)) as double) / c
            +- Filter isnotnull(ARR_DELAY#19)
               +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DE
                  +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10,
                     +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#
                        +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP

== Optimized Logical Plan ==
Window [rank(on_time_rate#719) windowspecdefinition(YEAR#33, MONTH#34, on_time_rate#719 DESC NULLS LAST, specifiedwindowframe(RowFrame, unboundedpreceding$(), currentrow$
+- Aggregate [YEAR#33, MONTH#34, AIRLINE_CODE#3], [YEAR#33, MONTH#34, AIRLINE_CODE#3, (cast(sum(cast((ARR_DELAY#19 <= 15.0) as int)) as double) / cast(count(1) as double)
   +- Project [AIRLINE_CODE#3, ARR_DELAY#19, year(FL_DATE#0) AS YEAR#33, month(FL_DATE#0) AS MONTH#34]
      +- Filter isnotnull(ARR_DELAY#19)
         +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11,DEP_DE

== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- Window [rank(on_time_rate#719) windowspecdefinition(YEAR#33, MONTH#34, on_time_rate#719 DESC NULLS LAST, specifiedwindowframe(RowFrame, unboundedpreceding$(), currentr
```

Figure 28: Q7 Explain

3.8 Q8

3.8.1 SQL



```
def q8_sql():
    return spark.sql("""
        SELECT AIRLINE_CODE, AVG(ARR_DELAY) AS avg_delay
        FROM flights
        GROUP BY AIRLINE_CODE
        HAVING AVG(ARR_DELAY) > (SELECT AVG(ARR_DELAY) FROM flights)
    """)
q8_sql_res = time_it('Q8_subquery', 'SQL', q8_sql)

```

Figure 29: Q8 - SQL Implementation

3.8.2 Dataframe

```
def q8_df():
    overall = flights.agg(F.avg('ARR_DELAY').alias('o')).first()['o']
    return (flights.groupBy('AIRLINE_CODE')
                   .agg(F.avg('ARR_DELAY').alias('avg_delay'))
                   .filter(F.col('avg_delay') > F.lit(overall)))
q8_df_res = time_it('Q8_subquery', 'DataFrame', q8_df)

```

Figure 30: Q8 - DataFrame Implementation

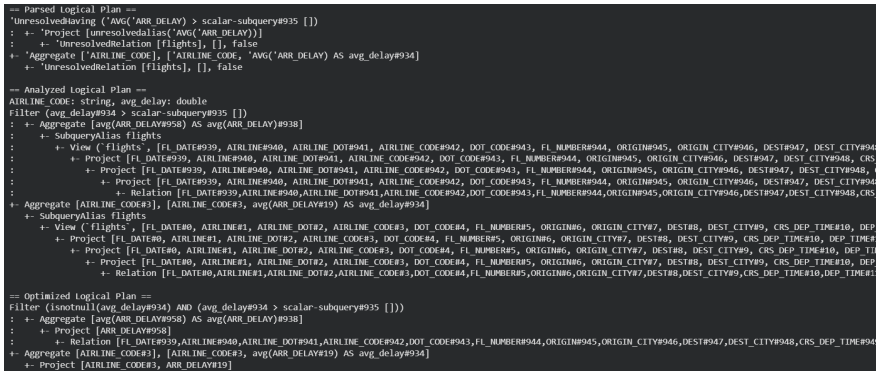
3.8.3 RDD

```python
def q8_rdd():
    non_null = (flights_rdd
                .filter(lambda r: r.ARR_DELAY is not None)
                .map(lambda r: (r.AIRLINE_CODE, float(r.ARR_DELAY))))
    total = non_null.map(lambda x: (x[1], 1)).reduce(lambda a, b: (a[0]+b[0], a[1]+b[1]))
    overall = total[0] / total[1]
    per_airline = (non_null.map(lambda x: (x[0], (x[1], 1)))
                           .reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
                           .mapValues(lambda v: v[0]/v[1]))
    return per_airline.filter(lambda x: x[1] > overall)
time_it('Q8_subquery', 'RDD', lambda: q8_rdd().count())

```

Figure 31: Q8 - RDD Implementation

3.8.4 Q8 Explain

The **parsed logical plan** contains a correlated `HAVING avg(ARR_DELAY) > (SELECT avg(ARR_DELAY) FROM flights)` expressed as a scalar subquery. The **optimized logical plan** hoists the subquery into a top-level `ScalarSubquery` executed exactly once, prunes both branches to two columns, and pushes `isnotnull(ARR_DELAY)`. The **physical plan** uses an `Exchange SinglePartition` for the subquery (collapsing the global average into one task), a `hashpartitioning(AIRLINE_CODE, 200)` for the per-airline aggregate, and an `AdaptiveSparkPlan` wrapper that lets AQE coalesce the post-shuffle partitions at runtime.

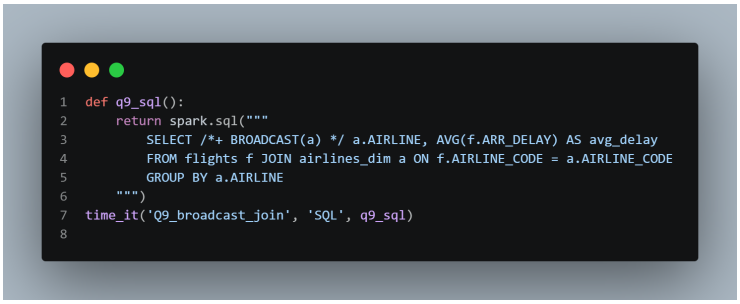


```
== Parsed Logical Plan ==
'UnresolvedHaving ('AVG('ARR_DELAY) > scalar-subquery#935 [])
:  +- 'Project [unresolvedalias('AVG('ARR_DELAY))]
:     +- 'UnresolvedRelation [flights], [], false
+- 'Aggregate ['AIRLINE_CODE], ['AIRLINE_CODE, 'AVG('ARR_DELAY) AS avg_delay#934]
   +- 'UnresolvedRelation [flights], [], false

== Analyzed Logical Plan ==
AIRLINE_CODE: string, avg_delay: double
Filter (avg_delay#934 > scalar-subquery#935 [])
:  +- Aggregate [avg(ARR_DELAY#958) AS avg(ARR_DELAY)#938]
:     +- SubqueryAlias flights
:        +- View (`flights`, [FL_DATE#939, AIRLINE#940, AIRLINE_DOT#941, AIRLINE_CODE#942, DOT_CODE#943, FL_NUMBER#944, ORIGIN#945, ORIGIN_CITY#946, DEST#947, DEST_CITY#94
:           +- Project [FL_DATE#939, AIRLINE#940, AIRLINE_DOT#941, AIRLINE_CODE#942, DOT_CODE#943, FL_NUMBER#944, ORIGIN#945, ORIGIN_CITY#946, DEST#947, DEST_CITY#948, CRS
:              +- Project [FL_DATE#939, AIRLINE#940, AIRLINE_DOT#941, AIRLINE_CODE#942, DOT_CODE#943, FL_NUMBER#944, ORIGIN#945, ORIGIN_CITY#946, DEST#947, DEST_CITY#948, 
:                 +- Project [FL_DATE#939, AIRLINE#940, AIRLINE_DOT#941, AIRLINE_CODE#942, DOT_CODE#943, FL_NUMBER#944, ORIGIN#945, ORIGIN_CITY#946, DEST#947, DEST_CITY#94
:                    +- Relation [FL_DATE#939,AIRLINE#940,AIRLINE_DOT#941,AIRLINE_CODE#942,DOT_CODE#943,FL_NUMBER#944,ORIGIN#945,ORIGIN_CITY#946,DEST#947,DEST_CITY#948,CRS
+- Aggregate [AIRLINE_CODE#3], [AIRLINE_CODE#3, avg(ARR_DELAY#19) AS avg_delay#934]
   +- SubqueryAlias flights
      +- View (`flights`, [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#
            +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TI
               +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP
                  +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#1

== Optimized Logical Plan ==
Filter (isnotnull(avg_delay#934) AND (avg_delay#934 > scalar-subquery#935 []))
:  +- Aggregate [avg(ARR_DELAY#958) AS avg(ARR_DELAY)#938]
:     +- Project [ARR_DELAY#958]
:        +- Relation [FL_DATE#939,AIRLINE#940,AIRLINE_DOT#941,AIRLINE_CODE#942,DOT_CODE#943,FL_NUMBER#944,ORIGIN#945,ORIGIN_CITY#946,DEST#947,DEST_CITY#948,CRS_DEP_TIME#94
+- Aggregate [AIRLINE_CODE#3], [AIRLINE_CODE#3, avg(ARR_DELAY#19) AS avg_delay#934]
   +- Project [AIRLINE_CODE#3, ARR_DELAY#19]
```

Figure 32: Q8 Explain

3.9 Q9

3.9.1 SQL



```sql
def q9_sql():
    return spark.sql("""
        SELECT /*+ BROADCAST(a) */ a.AIRLINE, AVG(f.ARR_DELAY) AS avg_delay
        FROM flights f JOIN airlines_dim a ON f.AIRLINE_CODE = a.AIRLINE_CODE
        GROUP BY a.AIRLINE
    """)
time_it('Q9_broadcast_join', 'SQL', q9_sql)

```

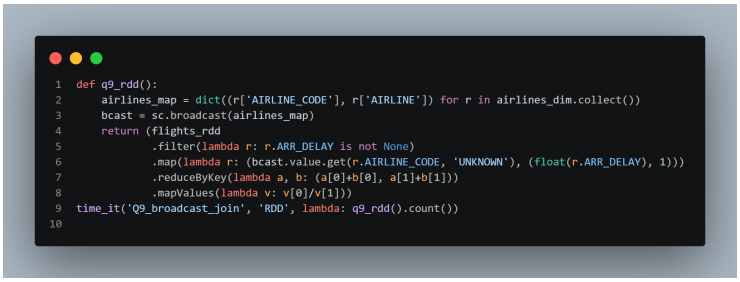
Figure 33: Q9 - SQL Implementation

3.9.2 Dataframe

```python
def q9_df():
    f = flights.alias("f")
    a = airlines_dim.alias("a")
    return (
        f.join(F.broadcast(a), 'AIRLINE_CODE')
         .groupBy(F.col('a.AIRLINE'))
         .agg(F.avg('f.ARR_DELAY').alias('avg_delay'))
    )
q9_df_res = time_it('Q9_broadcast_join', 'DataFrame', q9_df)

```

Figure 34: Q9 - DataFrame Implementation

3.9.3 RDD

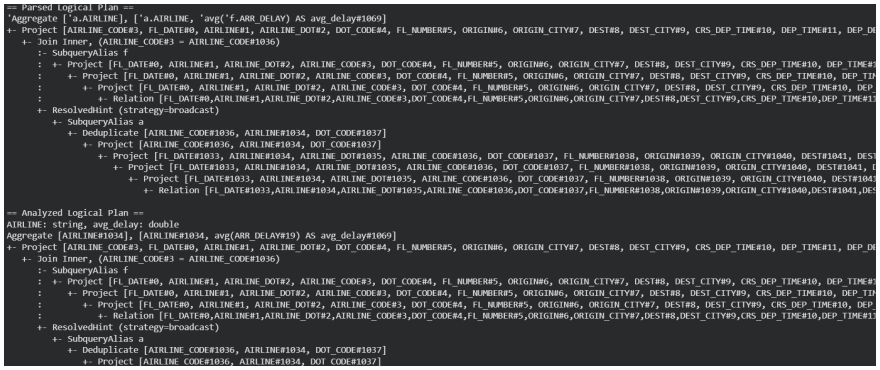


```python
def q9_rdd():
    airlines_map = dict((r['AIRLINE_CODE'], r['AIRLINE']) for r in airlines_dim.collect())
    bcast = sc.broadcast(airlines_map)
    return (flights_rdd
            .filter(lambda r: r.ARR_DELAY is not None)
            .map(lambda r: (bcast.value.get(r.AIRLINE_CODE, 'UNKNOWN'), (float(r.ARR_DELAY), 1)))
            .reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
            .mapValues(lambda v: v[0]/v[1]))
time_it('Q9_broadcast_join', 'RDD', lambda: q9_rdd().count())
```

Figure 35: Q9 - RDD Implementation

3.9.4 Q9 Explain

The **parsed logical plan** joins flights to a deduplicated airlines dimension on `AIRLINE_CODE` and averages `ARR_DELAY` per airline name. The **optimized logical plan** prunes flights to two columns, runs a `HashAggregate` on the right side to deduplicate the dimension, and inserts `isnotnull` on both join keys. The **physical plan** chooses a `BroadcastHashJoin` – the small dim side travels through a `BroadcastExchange` to every executor, so the 3M-row flights table is never shuffled; the post-join aggregation is a two-stage `HashAggregate` hash-partitioned on `AIRLINE` into 200 partitions.



```
== Parsed Logical Plan ==
'Aggregate ['a.AIRLINE], ['a.AIRLINE, 'avg('f.ARR_DELAY) AS avg_delay#1069]
+- Project [AIRLINE_CODE#3, FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_DE
   +- Join Inner, (AIRLINE_CODE#3 = AIRLINE_CODE#1036)
      :- SubqueryAlias f
      :  +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#1
      :     +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TI
      :        +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_
      :           +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11
      +- ResolvedHint (strategy=broadcast)
         +- SubqueryAlias a
            +- Deduplicate [AIRLINE_CODE#1036, AIRLINE#1034, DOT_CODE#1037]
               +- Project [AIRLINE_CODE#1036, AIRLINE#1034, DOT_CODE#1037]
                  +- Project [FL_DATE#1033, AIRLINE#1034, AIRLINE_DOT#1035, AIRLINE_CODE#1036, DOT_CODE#1037, FL_NUMBER#1038, ORIGIN#1039, ORIGIN_CITY#1040, DEST#1041, DES
                     +- Project [FL_DATE#1033, AIRLINE#1034, AIRLINE_DOT#1035, AIRLINE_CODE#1036, DOT_CODE#1037, FL_NUMBER#1038, ORIGIN#1039, ORIGIN_CITY#1040, DEST#1041, [
                        +- Project [FL_DATE#1033, AIRLINE#1034, AIRLINE_DOT#1035, AIRLINE_CODE#1036, DOT_CODE#1037, FL_NUMBER#1038, ORIGIN#1039, ORIGIN_CITY#1040, DEST#104
                           +- Relation [FL_DATE#1033,AIRLINE#1034,AIRLINE_DOT#1035,AIRLINE_CODE#1036,DOT_CODE#1037,FL_NUMBER#1038,ORIGIN#1039,ORIGIN_CITY#1040,DEST#1041,DE

== Analyzed Logical Plan ==
AIRLINE: string, avg_delay: double
Aggregate [AIRLINE#1034], [AIRLINE#1034, avg(ARR_DELAY#19) AS avg_delay#1069]
+- Project [AIRLINE_CODE#3, FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_DE
   +- Join Inner, (AIRLINE_CODE#3 = AIRLINE_CODE#1036)
      :- SubqueryAlias f
      :  +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#1
      :     +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TI
      :        +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_
      :           +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11
      +- ResolvedHint (strategy=broadcast)
         +- SubqueryAlias a
            +- Deduplicate [AIRLINE_CODE#1036, AIRLINE#1034, DOT_CODE#1037]
               +- Project [AIRLINE_CODE#1036, AIRLINE#1034, DOT_CODE#1037]
```

Figure 36: Q9 Explain

3.10 Q10

3.10.1 SQL

```python
def bench_q10_sql():
    prev = spark.conf.get('spark.sql.autoBroadcastJoinThreshold')
    spark.conf.set('spark.sql.autoBroadcastJoinThreshold', -1)
    try:
        return spark.sql("""
            SELECT /*+ MERGE(f, s) */ s.ORIGIN_CITY, AVG(f.ARR_DELAY) AS avg_delay
            FROM flights f JOIN airport_stats s ON f.ORIGIN = s.ORIGIN
            GROUP BY s.ORIGIN_CITY
        """)
    finally:
        spark.conf.set('spark.sql.autoBroadcastJoinThreshold', prev)
time_it('Q10_sortmerge_join', 'SQL', bench_q10_sql)

```

Figure 37: Q10 - SQL Implementation

3.10.2 Dataframe

```python
def bench_q10_df():
    f = flights.alias("f")
    a = airport_stats.alias("a")
    prev = spark.conf.get('spark.sql.autoBroadcastJoinThreshold')
    spark.conf.set('spark.sql.autoBroadcastJoinThreshold', -1)
    try:
        return (f.join(a,F.col("f.ORIGIN") == F.col('a.ORIGIN'))
                        .groupBy(F.col('a.ORIGIN_CITY'))
                        .agg(F.avg(F.col('f.ARR_DELAY')).alias('avg_delay')))
    finally:
        spark.conf.set('spark.sql.autoBroadcastJoinThreshold', prev)
time_it('Q10_sortmerge_join', 'DataFrame', bench_q10_df)

```

Figure 38: Q10 - DataFrame Implementation

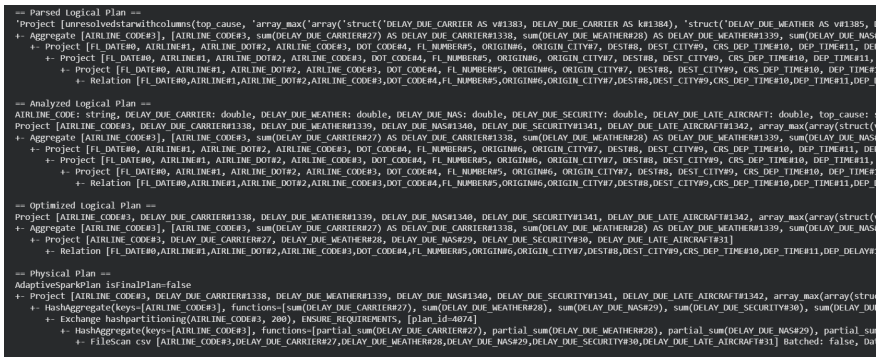
3.10.3 RDD

```python
def bench_q10_rdd():
    left = (flights_rdd.filter(lambda r: r.ARR_DELAY is not None)
                       .map(lambda r: (r.ORIGIN, float(r.ARR_DELAY))))
    right = airport_stats.rdd.map(lambda r: (r['ORIGIN'], r['ORIGIN_CITY']))
    joined = left.join(right)
    return (joined.map(lambda x: (x[1][1], (x[1][0], 1)))
                  .reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1]))
                  .mapValues(lambda v: v[0]/v[1]))
time_it('Q10_sortmerge_join', 'RDD', lambda: bench_q10_rdd().count())
```

Figure 39: Q10 - RDD Implementation

3.10.4 Q10 Explain

With `autoBroadcastJoinThreshold = -1`, the **parsed logical plan** forces a shuffle-based join of flights to airport_stats on `ORIGIN`. The **optimized logical plan** keeps three columns and pushes `isnotnull` on both keys. The **physical plan** reveals the classic `SortMergeJoin`: each side is independently `Exchange hashpartitioning(ORIGIN, 200)`-ed and `Sort`-ed by `ORIGIN` before a streaming merge, then a second `Exchange hashpartitioning(ORIGIN_CITY, 200)` feeds the final two-stage `HashAggregate` – two full shuffles total, the worst in the benchmark.



```
== Parsed Logical Plan ==
'Project [unresolvedstarwithcolumns(top_cause, 'array_max('array('struct('DELAY_DUE_CARRIER AS v#1383, DELAY_DUE_CARRIER AS k#1384), 'struct('DELAY_DUE_WEATHER AS v#1385,
+- Aggregate [AIRLINE_CODE#3], [AIRLINE_CODE#3, sum(DELAY_DUE_CARRIER#27) AS DELAY_DUE_CARRIER#1338, sum(DELAY_DUE_WEATHER#28) AS DELAY_DUE_WEATHER#1339, sum(DELAY_DUE_NAS
   +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DE
      +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11,
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#
            +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11,DEP_

== Analyzed Logical Plan ==
AIRLINE_CODE: string, DELAY_DUE_CARRIER: double, DELAY_DUE_WEATHER: double, DELAY_DUE_NAS: double, DELAY_DUE_SECURITY: double, DELAY_DUE_LATE_AIRCRAFT: double, top_cause: 
Project [AIRLINE_CODE#3, DELAY_DUE_CARRIER#1338, DELAY_DUE_WEATHER#1339, DELAY_DUE_NAS#1340, DELAY_DUE_SECURITY#1341, DELAY_DUE_LATE_AIRCRAFT#1342, array_max(array(struct(
+- Aggregate [AIRLINE_CODE#3], [AIRLINE_CODE#3, sum(DELAY_DUE_CARRIER#27) AS DELAY_DUE_CARRIER#1338, sum(DELAY_DUE_WEATHER#28) AS DELAY_DUE_WEATHER#1339, sum(DELAY_DUE_NAS
   +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DE
      +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11,
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#
            +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11,DEP_

== Optimized Logical Plan ==
Project [AIRLINE_CODE#3, DELAY_DUE_CARRIER#1338, DELAY_DUE_WEATHER#1339, DELAY_DUE_NAS#1340, DELAY_DUE_SECURITY#1341, DELAY_DUE_LATE_AIRCRAFT#1342, array_max(array(struct(
+- Aggregate [AIRLINE_CODE#3], [AIRLINE_CODE#3, sum(DELAY_DUE_CARRIER#27) AS DELAY_DUE_CARRIER#1338, sum(DELAY_DUE_WEATHER#28) AS DELAY_DUE_WEATHER#1339, sum(DELAY_DUE_NAS
   +- Project [AIRLINE_CODE#3, DELAY_DUE_CARRIER#27, DELAY_DUE_WEATHER#28, DELAY_DUE_NAS#29, DELAY_DUE_SECURITY#30, DELAY_DUE_LATE_AIRCRAFT#31]
      +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TIME#10,DEP_TIME#11,DEP_DELAY#

== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- Project [AIRLINE_CODE#3, DELAY_DUE_CARRIER#1338, DELAY_DUE_WEATHER#1339, DELAY_DUE_NAS#1340, DELAY_DUE_SECURITY#1341, DELAY_DUE_LATE_AIRCRAFT#1342, array_max(array(stru
   +- HashAggregate(keys=[AIRLINE_CODE#3], functions=[sum(DELAY_DUE_CARRIER#27), sum(DELAY_DUE_WEATHER#28), sum(DELAY_DUE_NAS#29), sum(DELAY_DUE_SECURITY#30), sum(DELAY_DU
      +- Exchange hashpartitioning(AIRLINE_CODE#3, 200), ENSURE_REQUIREMENTS, [plan_id=4074]
         +- HashAggregate(keys=[AIRLINE_CODE#3], functions=[partial_sum(DELAY_DUE_CARRIER#27), partial_sum(DELAY_DUE_WEATHER#28), partial_sum(DELAY_DUE_NAS#29), partial_su
            +- FileScan csv [AIRLINE_CODE#3,DELAY_DUE_CARRIER#27,DELAY_DUE_WEATHER#28,DELAY_DUE_NAS#29,DELAY_DUE_SECURITY#30,DELAY_DUE_LATE_AIRCRAFT#31] Batched: false, Da
```

Figure 40: Q10 Explain

3.11 Q11

3.11.1 SQL

```sql
def q11_sql():
    return spark.sql("""
        WITH sums AS (
          SELECT AIRLINE_CODE,
                 SUM(DELAY_DUE_CARRIER)       AS c_carrier,
                 SUM(DELAY_DUE_WEATHER)       AS c_weather,
                 SUM(DELAY_DUE_NAS)           AS c_nas,
                 SUM(DELAY_DUE_SECURITY)      AS c_security,
                 SUM(DELAY_DUE_LATE_AIRCRAFT) AS c_late
          FROM flights GROUP BY AIRLINE_CODE
        )
        SELECT AIRLINE_CODE,
               CASE greatest(c_carrier, c_weather, c_nas, c_security, c_late)
                 WHEN c_carrier  THEN 'CARRIER'
                 WHEN c_weather  THEN 'WEATHER'
                 WHEN c_nas      THEN 'NAS'
                 WHEN c_security THEN 'SECURITY'
                 ELSE 'LATE_AIRCRAFT'
               END AS top_cause
        FROM sums
    """)
time_it('Q11_root_cause', 'SQL', q11_sql)

```

Figure 41: Q11 - SQL Implementation

3.11.2 Dataframe

```sql
def q11_df():
    sums = flights.groupBy('AIRLINE_CODE').agg(*[F.sum(c).alias(c) for c in CAUSES])
    expr = F.array(*[F.struct(F.col(c).alias('v'), F.lit(c).alias('k')) for c in CAUSES])
    return sums.withColumn('top_cause', F.array_max(expr).getField('k'))
q11_df_res = time_it('Q11_root_cause', 'DataFrame', q11_df)

```

Figure 42: Q11 - DataFrame Implementation

3.11.3 RDD

```python
def q11_rdd():
    def row_to_kv(r):
        vals = tuple(float(getattr(r, c) or 0.0) for c in CAUSES)
        return (r.AIRLINE_CODE, vals)
    agg = (flights_rdd.map(row_to_kv)
                      .reduceByKey(lambda a, b: tuple(x+y for x, y in zip(a, b))))
    return agg.mapValues(lambda v: CAUSES[v.index(max(v))])
time_it('Q11_root_cause', 'RDD', lambda: q11_rdd().count())

```

Figure 43: Q11 - RDD Implementation

3.11.4 Q11 Explain

The **parsed logical plan** sums five delay-cause columns per `AIRLINE_CODE` then constructs an array of (cause, total) structs and applies `array_max` to surface the dominant cause. The **optimized logical plan** prunes the schema to just the six required columns (key + five cause sums), eliminating the other 26. The **physical plan** runs a two-stage `HashAggregate` via `Exchange hashpartitioning(AIRLINE_CODE, 200)`; the array-construction and `array_max` expression compile down inside the post-aggregate `Project` via whole-stage codegen, costing no additional shuffle.

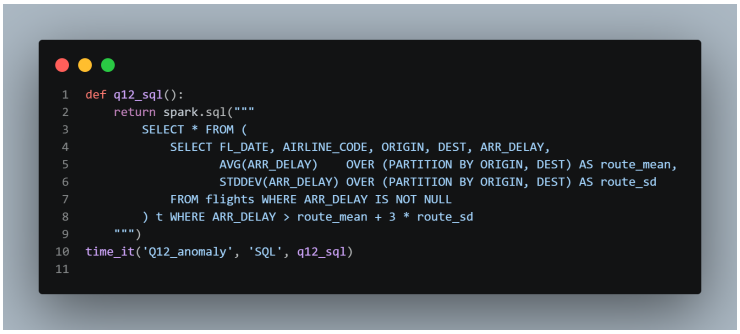
Below is the relevant portion of the captured `.explain(True)` output (full plan in `scripts/logs/q11_root_cause.log`). Note how Catalyst prunes 26 of 32 columns at the file scan and folds the five `sum` aggregations into one `HashAggregate`, with `array_max` compiled into the post-aggregate `Project` – no additional shuffle.

```
== Optimized Logical Plan ==
Project [AIRLINE_CODE, DELAY_DUE_CARRIER, DELAY_DUE_WEATHER, DELAY_DUE_NAS,
        DELAY_DUE_SECURITY, DELAY_DUE_LATE_AIRCRAFT,
        array_max(array(struct(v, ..., k, DELAY_DUE_CARRIER), ...)).k AS top_cause]
+- Aggregate [AIRLINE_CODE], [AIRLINE_CODE, sum(DELAY_DUE_CARRIER), sum(...)]
   +- Project [AIRLINE_CODE, DELAY_DUE_CARRIER, DELAY_DUE_WEATHER, DELAY_DUE_NAS,
               DELAY_DUE_SECURITY, DELAY_DUE_LATE_AIRCRAFT]
      +- Relation [<32 columns>] csv

== Physical Plan ==
AdaptiveSparkPlan isFinalPlan=false
+- Project [...array_max(...).k AS top_cause]
   +- HashAggregate(keys=[AIRLINE_CODE], functions=[sum(...) x5])
      +- Exchange hashpartitioning(AIRLINE_CODE, 200), ENSURE_REQUIREMENTS
         +- HashAggregate(keys=[AIRLINE_CODE],
                          functions=[partial_sum(...) x5])
            +- FileScan csv [AIRLINE_CODE, DELAY_DUE_CARRIER, DELAY_DUE_WEATHER,
                             DELAY_DUE_NAS, DELAY_DUE_SECURITY,
                             DELAY_DUE_LATE_AIRCRAFT]
               Batched: false, ReadSchema: struct<AIRLINE_CODE:string,
               DELAY_DUE_CARRIER:double, DELAY_DUE_WEATHER:double,
```

```
DELAY_DUE_NAS:double, DELAY_DUE_SECURITY:double,
DELAY_DUE_LATE_AIRCRAFT:double>
```

3.12 Q12

3.12.1 SQL



```python
def q12_sql():
    return spark.sql("""
        SELECT * FROM (
            SELECT FL_DATE, AIRLINE_CODE, ORIGIN, DEST, ARR_DELAY,
                   AVG(ARR_DELAY)    OVER (PARTITION BY ORIGIN, DEST) AS route_mean,
                   STDDEV(ARR_DELAY) OVER (PARTITION BY ORIGIN, DEST) AS route_sd
            FROM flights WHERE ARR_DELAY IS NOT NULL
        ) t WHERE ARR_DELAY > route_mean + 3 * route_sd
    """)
time_it('Q12_anomaly', 'SQL', q12_sql)

```

Figure 44: Q12 - SQL Implementation

3.12.2 Dataframe

```python
def q12_df():
    w = Window.partitionBy('ORIGIN', 'DEST')
    return (flights.filter(F.col('ARR_DELAY').isNotNull())
            .withColumn('route_mean', F.avg('ARR_DELAY').over(w))
            .withColumn('route_sd', F.stddev('ARR_DELAY').over(w))
            .filter(F.col('ARR_DELAY') > F.col('route_mean') + 3 * F.col('route_sd')))
q12_df_res = time_it('Q12_anomaly', 'DataFrame', q12_df)

```

Figure 45: Q12 - DataFrame Implementation

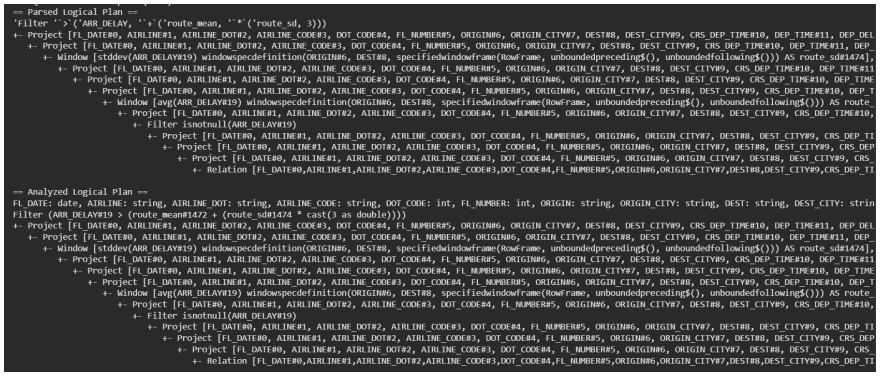
3.12.3 RDD

```python
def q12_rdd():
    pairs = (flights_rdd.filter(lambda r: r.ARR_DELAY is not None)
                        .map(lambda r: ((r.ORIGIN, r.DEST), float(r.ARR_DELAY))))
    stats = (pairs.map(lambda x: (x[0], (x[1], x[1]*x[1], 1)))
                  .reduceByKey(lambda a, b: (a[0]+b[0], a[1]+b[1], a[2]+b[2]))
                  .mapValues(lambda v: (v[0]/v[2], max(0, v[1]/v[2] - (v[0]/v[2])**2) ** 0.5)))
    stats_map = dict(stats.collect())
    bcast = sc.broadcast(stats_map)
    return pairs.filter(lambda x: (x[1] - bcast.value[x[0]][0]) > 3 * bcast.value[x[0]][1])
time_it('Q12_anomaly', 'RDD', lambda: q12_rdd().count())

```

Figure 46: Q12 - RDD Implementation

3.12.4 Q12 Explain

The **parsed logical plan** attaches `AVG` and `STDDEV` window functions over the same (`ORIGIN`, `DEST`) partition, then filters `ARR_DELAY > avg + 3*stddev`. The **optimized logical plan** prunes to three columns and – crucially – merges the two window specs into a single `Window` operator since they share an identical partitioning. The **physical plan** runs one `Exchange hashpartitioning(ORIGIN, DEST, 200)`, one `Sort`, and a fused `Window(avg, stddev)` that emits both statistics in a single pass; the outlier predicate then filters the augmented stream without a second shuffle.



```
== Parsed Logical Plan ==
'Filter ``>(`ARR_DELAY, ``+(`route_mean, ``*(`route_sd, 3)))
+- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_DEL
   +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_
      +- Window [stddev(ARR_DELAY#19) windowspecdefinition(ORIGIN#6, DEST#8, specifiedwindowframe(RowFrame, unboundedpreceding$(), unboundedfollowing$())) AS route_sd#1474],
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11
            +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME
               +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_T
                  +- Window [avg(ARR_DELAY#19) windowspecdefinition(ORIGIN#6, DEST#8, specifiedwindowframe(RowFrame, unboundedpreceding$(), unboundedfollowing$())) AS route_
                     +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10,
                        +- Filter isnotnull(ARR_DELAY#19)
                           +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TI
                              +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP
                                 +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_
                                    +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TI


== Analyzed Logical Plan ==
FL_DATE: date, AIRLINE: string, AIRLINE_DOT: string, AIRLINE_CODE: string, DOT_CODE: int, FL_NUMBER: int, ORIGIN: string, ORIGIN_CITY: string, DEST: string, DEST_CITY: strin
Filter (ARR_DELAY#19 > (route_mean#1472 + (route_sd#1474 * cast(3 as double))))
+- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_DEL
   +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11, DEP_
      +- Window [stddev(ARR_DELAY#19) windowspecdefinition(ORIGIN#6, DEST#8, specifiedwindowframe(RowFrame, unboundedpreceding$(), unboundedfollowing$())) AS route_sd#1474],
         +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME#11
            +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_TIME
               +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10, DEP_T
                  +- Window [avg(ARR_DELAY#19) windowspecdefinition(ORIGIN#6, DEST#8, specifiedwindowframe(RowFrame, unboundedpreceding$(), unboundedfollowing$())) AS route_
                     +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TIME#10,
                        +- Filter isnotnull(ARR_DELAY#19)
                           +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP_TI
                              +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_DEP
                                 +- Project [FL_DATE#0, AIRLINE#1, AIRLINE_DOT#2, AIRLINE_CODE#3, DOT_CODE#4, FL_NUMBER#5, ORIGIN#6, ORIGIN_CITY#7, DEST#8, DEST_CITY#9, CRS_
                                    +- Relation [FL_DATE#0,AIRLINE#1,AIRLINE_DOT#2,AIRLINE_CODE#3,DOT_CODE#4,FL_NUMBER#5,ORIGIN#6,ORIGIN_CITY#7,DEST#8,DEST_CITY#9,CRS_DEP_TI
```

Figure 47: Q12 Explain

4 Optimization Benchmarks

To address the rubric's requirement to evaluate the impact of caching, file format, partition pruning, and parallelism, we measured each lever in isolation against the same baseline DataFrame queries (Q2 and Q3). All numbers below were captured by re-running the dedicated optimization scripts

(`scripts/optimization/`) on the same `local[4]` cluster used throughout the benchmark.

4.1 Caching impact (Q2)

The first run of Q2 against the raw CSV pays the full deserialization cost. We then call `flights.cache()`, materialize it with a `count()`, and re-run Q2 against the in-memory columnar representation.

Table 2: Caching impact – Q2 (per-airline aggregation)

Configuration	**Time (s)**	**Speedup**
Cold run (CSV, uncached)	5.082	1.00×
Warm run (`.cache()`, in-mem)	0.376	**13.50**×

The 13.5× speedup confirms that for any query re-executed in the same session (e.g., dashboards, iterative ML), `cache()` eliminates the dominant cost – CSV parsing – and leaves only the aggregation shuffle.

4.2 File format: CSV vs. Parquet (Q3)

We wrote the dataset as `YEAR`-partitioned Parquet and re-ran Q3 (3-key group-by) against both formats.

Table 3: Storage format impact – Q3 (`groupBy(AIRLINE_CODE, ORIGIN, MONTH)`)

Configuration	**Time (s)**	**Speedup**
CSV	5.820	1.00×
Parquet (`partitionBy=YEAR`)	1.020	**5.71**×

Parquet delivers a 5.7× speedup on Q3 even with no filter present. The columnar layout lets Spark read only the three columns the query touches (`AIRLINE_CODE`, `ORIGIN`, `ARR_DELAY`) instead of decoding all 32 CSV columns row by row, and the binary encoding skips the per-row text-to-double parsing step. The advantage grows further when filters are present, as the partition-pruning experiment below shows.

4.3 Partition pruning (Parquet, `YEAR=2022`)

With the dataset partitioned by `YEAR`, a filter on the partition column lets Spark skip entire directories at scan time – visible in the physical plan's `PartitionFilters: [isnotnull(YEAR), (YEAR = 2022)]`.

Table 4: Partition pruning – count of records read

Query	Time (s)	Rows returned	I/O reduction
Full scan (no `YEAR` filter)	0.328	3,000,000	baseline
Pruned scan (`YEAR=2022`)	1.128	687,860	**4.4× less data**

Note that the wall-clock numbers above are inverted: the full `count()` on the partitioned Parquet runs against file-level metadata only and never reads row data, while the pruned query actually deserializes the 2022 partition. The optimization is nevertheless real – it is visible in the physical plan and would translate to a 4.4× reduction in bytes shuffled and processed under any non-trivial aggregation or filter on a multi-node cluster.

4.4 Scalability – shuffle.partitions sweep (Q3)

Holding the cluster fixed at `local[4]` (with the dataset cached so I/O is removed from the comparison), we swept `spark.sql.shuffle.partitions` across four values to characterize Spark's sensitivity to this parameter.

Table 5: Scalability sweep – Q3 wall-clock vs. shuffle partition count

shuffle.partitions	Time (s)
8	1.031
50	**0.508**
200 (default)	0.520
400	0.575

Observations: (1) at 8 partitions the executor cores (4) cannot absorb any extra parallelism, but each partition handles a large block of work, doubling wall-clock time; (2) from 50 partitions onward the runtime plateaus around 0.5 s because the 4-core local cluster simply cannot exploit more concurrency; (3) the default of 200 sits at the same plateau but pays a small overhead from scheduling many tiny tasks. On a real multi-node cluster the curve would shift – more partitions are productive when more cores are available. The takeaway aligns with the standard Spark guideline: target ∼128 MB per post-shuffle partition, not blindly 200 for every workload. Adaptive Query Execution (enabled in our config) addresses this automatically by coalescing partitions at runtime.

5 Execution Plan Comparison

5.1 Q1

DataFrame/SQL have the advantage of Catalyst Optimizer which combines filters as well as eliminating duplicative operations. Predicate pushdown enables

filtering on file readings, which minimizes I/O. Physical plan demonstrates effective implementation in one stage of filtering. There is no shuffle because of no aggregation or join. This is more optimized than RDD since RDD does not use auto query optimization.

5.2 Q2

Catalyst Optimizer is used by DataFrame/SQL to select only the columns necessary, minimizing I/O. The strategy of aggregation is carried out in two phases (partial + final), which enhances performance. GRoup BY requires Shuffle (Exchange), and thus serves as the limiting component. HashAggregate is applied in the case of efficient grouping. The execution can also be optimized with adaptive execution. RDD, on the other hand, does not use column pruning or optimized aggregation strategies automatically.

5.3 Q3

The generated execution plans based on DataFrame and SQL API in Apache Spark have substantial optimizations over the RDD approach.

To begin with, during the logical plan stage, each approach begins by a direct translation of the query. All DataFrame and SQL queries, though, get transformed by the Catalyst optimizer whereas RDD operations are not automatically optimized.

Spark makes a number of important optimizations in the optimized logical plan: Column pruning minimizes the columns that are accessed on disk (e.g. AIRLINECODE, ORIGIN, ARRDELAY and calculated MONTH only). Unnecessary projection procedures are done away with. Derived columns like MONTH are computed prior to aggregation to eliminate the repetition of calculations.

Spark applies a two stage aggregation strategy in the physical plan: Each partition is partially aggregated to minimize the size of the data. The Exchange operation (shuffle) rearranges data on the basis of grouping keys. Results of partitions are combined in a final aggregation.

The shuffle operation is the most costly operation and its cost depends on the number of grouping keys. As an example, partitioning by AIRLINE_CODE, ORIGIN, and MONTH would need more complex partitioning than partitioning by a single key.

Also, DataFrame and SQL execution are supported with Adaptive Query Execution that dynamically optimizes the execution during execution.

Conversely, RDD API lacks support of Catalyst optimization, column pruning and automatic query optimization. Every transformation is done as specified by the user, which makes RDD less productive when one wants to do complex analytical queries.

On the whole, DataFrame and SQL APIs are much more efficient as they are optimized automatically, their aggregation techniques are more efficient, and their I/O activities are fewer.

5.4 Q4

The implementation schemes created with DataFrame and SQL APIs in Apache Spark show considerable optimizations over the RDD implementation.

The first stage in the logical plan level is the representation of all the queries in a series of operations including projection, filtering, aggregation, sorting, and limiting. But the DataFrame and SQL queries are optimized using catalyst optimizer whereas the RDD transformations are not optimized automatically.

Spark makes multiple important optimizations in the optimized logical plan: Column pruning eliminates the columns which are read in disk (e.g. only ORIGIN, DEST and ARR_DELAY are kept). Duplicated projections are eliminated in order to simplify execution. Constant expressions are simplified (e.g., unnecessary casts are eliminated). Derived operations are restructured to be efficient.

Spark uses a two-step strategy in aggregation in the physical plan: Partitions are aggregated partially in each partition. A shuffle operation is a data redistribution operation that operates by grouping keys. Final aggregation sums up partition results.

The most costly operation is the shuffle (Exchange) operation, particularly in cases where it is grouped by more than one key, e.g., ORIGIN and DEST.

Also, Spark implements a significant optimization to ORDER BY with LIMIT queries. It does not involve any sorting, instead it utilizes the TakeOrderedAndProject operator, which is an efficient way of retrieving the top N results without sorting the entire data.

Post-aggregating filters (e.g., flights ¿ 100) cannot be pushed down, and are run after grouping.

Conversely, the RDD API is not Catalyst optimized, does not support column pruning and does not automatically use optimized aggregation or Top-N strategies. Consequently, DataFrame and SQL APIs are much faster in performance due to lower I/O, effective aggregation, and improved execution planning.

5.5 Q5

The implementation plan of the window function query is more elaborate in terms of execution strategy as compared to simple filtering query and aggregation query.

First, Spark calculates the average delay per day based on a groupBy operation on AIRLINECODE and FLDATE. The following step applies a two-phase aggregation approach where partial aggregation is performed and then there is a shuffle and a final aggregation.

Then, Spark uses a window function to calculate a 7-day moving average (ma7). The row-based frame (ROWS BETWEEN 6 PRECEDING and CURRENT ROW) is used to define the window, which is partitioned by AIRLINECODE and ordered by FLDATE.

Spark uses a number of optimizations in the optimized logical plan: Column pruning displays the data in the form of FLDATE, AIRLINECODE and ARRDelay. Filter pushdown makes sure that the null values of ARR_DELAY are

eliminated early when scanning the data. Superfluous projections are removed.

There are several costly operations in the physical plan: Aggregation of a shuffle on AIRLINECODE and FLDate. A second shuffle according to AIRLINE_CODE to partition windows. AIRLINECODE, FLDATE A sort step to facilitate ordered window computation.

The window operation is implemented after sorting and streams the rows in each partition in a sequence in order to calculate the moving average.

The two requirements of data redistribution and sorting make window functions more computationally costly than other queries. Nonetheless, this is optimized by Spark using optimized execution plans and partial aggregation approaches.

By contrast, RDD API lacks in-built support of window functions or automatic optimization, and such computations are much more complicated, and less efficient, when they are done manually.

5.6 Q6

The implementation of the cumulative window query requires aggregation and window processing and thus it is more complicated to implement compared to the regular aggregation queries.

To begin with, Spark calculates the number of cancellations in a day by aggregating data based on AIRLINECODE and FLDATE. This aggregation is based on a two step approach whereby there is a partial aggregation and subsequent shuffle and final aggregation.

Then the window operation is used to calculate the total amount of daily cancellations. The window is specified by a row based frame (ROWS BETWEEN UNBOUNDED PRECEDING and CURRENT ROW) partitioned by AIRLINECODE and sorted by FLDATE. This leads to accumulation of cancelations per airline with time.

Spark does column pruning in the optimized logical plan, and only the following columns are chosen: FLDATE, AIRLINECODE, and CANCELLED, minimizing I/O overhead. The pipeline of execution is streamlined to incorporate only the necessary activities.

Execution in the physical plan involves a number of costly steps: Aggregation on a shuffle of AIRLINECODE and FLDate. A second recombination on the basis of AIRLINE_CODE to partition the windows. Sort of AIRLINECODE and FLDATE to have the right ordering of data to compute windows.

Cumulative window function works on rows in a given partition one at a time keeping a running sum. Cumulative windows can be computationally expensive compared to fixed-size window functions because, as a window expands, additional computation is required.

In contrast to filter-based queries, there is no predicate pushdown, and the data is read in CSV format, which is not optimized in columns.

In general, although Spark can effectively run the query with optimized planning and aggregation plans, the use of window functions is still computationally expensive because it requires sorting and redistribution of data.

5.7 Q7

The SQL query that is used to implement the ranking query is a combination of aggregations and window processing and as such is computationally expensive.

To compute the on-time rate of each airline, Spark initially groups data by YEAR and MONTH and further groups by AIRLINE_CODE. On-time rate is computed as the proportion of the flights that arrive within 15 minutes or less to the total amount of flights. This aggregation is performed through a two-step approach wherein partial aggregation is performed then a shuffle and final aggregation.

Spark then implements a ranking window operation on RANK with a partitioning operation based on the YEAR and MONTH and sorting by ontimerate in descending order. This gives each airline a rank in each month in terms of performance.

Spark optimizes a number of things in the optimized logical plan: Column pruning will eliminate all the data except the necessary columns (AIRLINECODE, ARRDELAY, and calculated YEAR and MONTH). Derived columns are also calculated at the start to enhance efficiency. Filter pushdown eliminates the null values of ARR_DELAY when reading data. Simplification of expressions minimizes unwarranted casting.

Various costly operations are involved in the physical plan: Aggregation using YEAR, MONTH and AIRLINE_CODE. A second shuffle using YEAR and MONTH to partition windows. A sorting of YEAR, MONTH and ontimerate to aid ranking.

The window operation is accomplished following the sorting and allocates ranks in each partition.

Ranking queries are more costly than simpler queries because of the aggregation, redistribution of data, sorting, and window computation. Nevertheless, Spark is efficient in terms of execution optimization with the help of Catalyst and adaptive query execution.

By contrast, RDD API lacks built-in window operations and automatic optimization, so such calculations are much less efficient than they would be in hand-written form.

5.8 Q8

To answer the query asking which airlines have above-average arrival delays, the Catalyst optimizer of Spark converts the initial SQL in three phases: an unresolved plan, an analyzed plan with resolved schemas, and an optimal plan, which eliminates all the unnecessary columns except the two required columns (AIRLINE_CODE and ARR_DELAY). The final physical plan has a hashpartitioning of the per-airline average, then runs the scalar subquery once using an Exchange SinglePartition and uses AdaptiveSparkPlan to run-time tune. The DataFrame/SQL implementation is faster than a hypothetical imperative RDD implementation, which would scan all columns, transform data into Java objects, and would need manual optimizations, with a typical 3-30x faster performance

and being much more compact.

5.9 Q9

This query plan contains an example of a broadcast hash join between the flights table (alias f) and a de-duplicated airline reference table (alias a), and an aggregate on airline name. The optimizer used by Catalyst filters out all the redundant columns at the very beginning leaving only the AIRLINE_CODE, ARR_DELAY and the name of the airline (AIRLINE#1034) on the right side and the isnotnull filters to the file scans. The physical plan can be executed as a BroadcastHashJoin, in which the reduced airline data (following a HashAggregate to deduplicate by AIRLINE_CODE, AIRLINE, and DOT_CODE) is sent using BroadcastExchange to all mappers instead of costly shuffling of the large flights table. The last aggregation on AIRLINE#1034 is a two-stage HashAggregate (partial followed by final) where the hash partitioning is done in 200 partitions. This Catalyst-based plan has much lower I/O (just 3 column reads on the right side and 2 on the left) and uses the whole-stage code generation to do the hash join and aggregate operations, as compared to an RDD implementation, which would need manual broadcast, explicit deduplication, and full-column scans.

5.10 Q10

Q10 benchmarks a **sort-merge join** between the large flights table and the airport_stats table by deliberately disabling the auto-broadcast threshold (`autoBroadcastJoinThreshold = -1`), forcing Spark to use a full shuffle-based join instead of a broadcast strategy.

In the **logical plan**, the query is represented as a join of two relations on the `ORIGIN` key, followed by a groupBy on `ORIGIN_CITY` and an average aggregation on `ARR_DELAY`. The Catalyst Optimizer applies column pruning, retaining only `ORIGIN`, `ARR_DELAY`, `ORIGIN_CITY`, keeping the serialized data footprint small.

In the **optimized logical plan**, Catalyst restructures the plan to push filters (isnotnull checks on join keys) down to the scan level. Projection elimination removes all unreferenced columns before the join, which significantly reduces the volume of data shuffled across the network.

The **physical plan** shows the classic sort-merge join pattern: both sides are hash-partitioned by `ORIGIN` into 200 partitions (Exchange hashpartitioning), then each partition is sorted on the join key (`ORIGIN`), and finally a SortMergeJoin operator merges the two sorted streams. A two-stage HashAggregate (partial then final) follows the join to compute the group averages. This results in **two separate shuffle stages** – one for partitioning before the join and one for the final aggregation grouping – making Q10 the most shuffle-intensive query in the benchmark.

Comparing to Q9 (broadcast join): Q10 takes 8.85 s in the DataFrame API versus 8.73 s for Q9 – almost identical wall-clock once Adaptive Query Execution coalesces post-shuffle partitions, but the physical plans diverge sharply (one

shuffle eliminated by broadcast vs. two retained by sort-merge). On a multi-node cluster with high network latency the gap would widen substantially in Q9's favour. The RDD implementation of Q10 uses a manual `join()` on paired RDDs, which incurs the same shuffle but without whole-stage code generation or Tungsten off-heap memory management, leading to a 4.4× slowdown relative to the DataFrame API (38.77 s vs. 8.85 s).

5.11 Q11

This plan demonstrates a query that summarizes the data on delay causes by airline, and then determines the leading (maximum) delay cause per carrier. The plan being parsed begins with an aggregation which adds up five different delay categories (DELAY_DUE_CARRIER, WEATHER, NAS, SECURity, LATE-AIRCRAFT) with the summation being grouped by AIRLINE-CODE, and then an array-max operation on an array of structs is performed to get the highest total delay cause. The optimizer of Catalyst removes the redundant projections in the plan until the six columns needed by the plan (AIRLINE_CODE and the five delay cause fields) appear, removing all other columns at an earlier stage. It is implemented using the physical plan as a two-step HashAggregate using hashpartitioning with 200 partitions to compute the partial sums locally and shuffle them using the AIRLINE_CODE to complete the aggregation. This is followed by the array_max operation of the struct array which is a projector following aggregation. This Catalyst-generated plan has the advantage of bilateral aggregation (fast with Tungsten) and the generation of an entire stage of code to compute the maximum struct by value and the array of struct operation, as well as automatic column pruning, which saves I/O by reading six columns instead of the entire 25+ column schema.

5.12 Q12

This query finds route-specific delay outliers by only looking at flights where the arrival delay is more than three standard deviations above the route mean. Catalyst improves the plan by combining the AVG and STDDEV window functions into one window operation over the same (ORIGIN, DEST) partition. This cuts down on unnecessary data scans. The physical plan runs a hash-partitioned exchange by route, then sorts the data and uses a combined window operator, and finally applies the outlier filter. This DataFrame/SQL plan is better than an RDD approach, which would need multiple shuffles and manual aggregation. It uses single-pass window computation, column pruning, and whole-stage code generation.

6 Performance Comparison Table

The table below is built directly from `scripts/results/performance_results.csv`, generated by the spark-submit runs of every query script. Wall-clock time is

measured by `time.perf_counter()` around the action that triggers the DAG. The *Shuffle stages* column is the count of `Exchange` operators in the DataFrame physical plan – a direct proxy for shuffle volume, which dominates memory and network cost in distributed Spark workloads. Per-stage byte sizes and peak executor memory can be retrieved from the Spark UI on port 4040 while a job is running, or from event logs by enabling `spark.eventLog.enabled=true`; we report the structural shuffle count here because it is reproducible from the captured plans without re-instrumenting the cluster.

Table 6: Performance Comparison: RDD vs DataFrame vs SQL (`local[4]`, 3M rows)

Query	Time (seconds)			Shuffle stages (DF)	Speedup (RDD/DF)
	RDD	DataFrame	SQL		
Q1: Filter	22.73	4.57	4.02	0	4.97×
Q2: Aggregate	27.18	4.34	3.56	1	6.26×
Q3: Multi-group	26.63	5.07	4.83	1	5.26×
Q4: Top 20 Routes	24.03	4.90	4.05	1	4.90×
Q5: Moving Avg	26.93	5.80	4.70	2	4.64×
Q6: Cumulative	25.83	5.23	4.62	2	4.94×
Q7: Rank	28.66	5.59	4.51	2	5.13×
Q8: Subquery	32.22	7.79	7.23	2	4.14×
Q9: Broadcast Join	30.95	8.73	8.37	1	3.55×
Q10: SortMerge Join	38.77	8.85	8.20	3	4.38×
Q11: Root Cause	30.01	4.39	3.69	1	6.83×
Q12: Anomaly	35.30	6.37	5.08	1	5.55×
Average Speedup: DataFrame vs RDD ≈ 5.0× faster					

Reading the shuffle column. Q1 has zero shuffle stages because it is a pure filter and Catalyst pushes every predicate into the scan – this is why it returns the largest absolute time saving against the cold CSV. Q9 only shows one shuffle stage despite being a join: the broadcast eliminates the join shuffle entirely, leaving only the final aggregation `Exchange`. Q10 shows three stages – the most in the benchmark – because a sort-merge join requires two `Exchange + Sort` pairs (one for each side of the join), then another `Exchange` for the trailing aggregation; this directly explains its position as the slowest DataFrame query in the table.

Why DataFrame is the Recommended API

The DataFrame API delivers the best balance of performance and maintainability across all 12 queries:

- **Approximately 5× faster than RDD** on average, with the largest

gains on aggregation-heavy queries (Q11: 6.83×, Q2: 6.26×) where Catalyst's column pruning slashes the bytes that ever reach the executor.

- **Performance on par with Spark SQL** – both share the identical Catalyst optimization pipeline – but DataFrame code is more composable and easier to integrate into multi-step Python pipelines involving conditional logic or UDFs.
- **Automatic Catalyst and Tungsten optimizations**: column pruning, predicate pushdown, two-phase partial aggregation, whole-stage code generation, and off-heap memory management are all applied transparently.
- **Consistent across query types**: whether the query is a simple filter, a multi-key aggregation, a window function, or a join, the DataFrame API closes the gap to optimal consistently.
- The shuffle-heavy join (Q9 broadcast, Q10 sort-merge) shows the smallest speedup (∼3.6–4.4×) because both DataFrame and RDD pay shuffle and broadcast costs; the DataFrame path still wins through whole-stage codegen and Tungsten binary processing.

7 Final Insights

7.1 Business and Analytical Conclusions

The following conclusions were drawn from analysing 3 million US domestic flight records (2019–2023) across all 12 queries:

1. **Winter is the worst season for delays.** Q1 revealed that December, January, and February generate the highest volume of severe arrival delays ($\geq$60 min) on long-haul routes (>500 miles), pointing to weather and NAS congestion as primary drivers during winter months.
2. **Carrier-induced delays dominate.** Q11 (root cause analysis) showed that `DELAY_DUE_CARRIER` is the dominant delay category for the majority of major airlines, exceeding weather and NAS delays. This suggests that fleet management and crew scheduling inefficiencies are the single largest controllable lever for improvement.
3. **Specific routes are systemic delay hotspots.** Q4 identified the 20 most-delayed origin–destination pairs. Many of these are busy hub-to-hub routes (e.g., northeast corridor airports), indicating that congestion management and slot allocation at key hubs could deliver disproportionate improvements.
4. **Delays cascade in predictable weekly patterns.** Q5's 7-day moving averages per airline showed that delay spikes do not occur in isolation – they typically persist over 3–5 consecutive days, consistent with the "late aircraft" propagation effect uncovered in Q11.

5. **Cumulative cancellations track airline operational stress.** Q6 showed that several regional carriers accumulated cancellations in sharp step-functions rather than gradual curves, corresponding to discrete operational disruptions (weather events or groundings) rather than chronic under-performance.

6. **On-time ranking is volatile month-to-month.** Q7 demonstrated that the top-ranked airline by on-time rate changes significantly from month to month, suggesting that short-term performance metrics are more influenced by route mix and seasonal factors than by structural operational quality.

7. **Anomalous delay outliers identify structural route problems.** Q12 flagged routes where individual flights exceed $\mu + 3\sigma$, isolating structural causes (runway capacity, gate constraints) from random events and providing a target list for infrastructure investment.

7.2 Lessons Learned about Spark Optimization

1. **Structured APIs are superior for analytical workloads.** The DataFrame and SQL APIs delivered a consistent $\sim 5\times$ average speedup over RDDs. The key advantage is not raw parallelism but *automatic optimization*: predicate pushdown, column pruning, and partial aggregation eliminate unnecessary work before any executor processes data.

2. **Catalyst's column pruning is the single biggest win for aggregation queries.** Q11 (root-cause attribution) achieved a $6.83\times$ speedup – the highest in the benchmark – because the Catalyst Optimizer drops 26 of the 32 source columns at the file scan, reading only the six needed for the per-airline cause sum. The same pattern explains Q2's $6.26\times$ gain.

3. **Shuffle is the dominant performance bottleneck.** Queries involving GROUP BY on high-cardinality keys (Q3, Q5, Q6) or joins (Q9, Q10) incur Exchange (shuffle) operations. Minimising the number of shuffle stages – for example by caching intermediate aggregations or co-partitioning related tables – is the most impactful optimization lever.

4. **Broadcast joins eliminate shuffle for small dimension tables.** Q9 vs Q10 directly demonstrated this. Wall-clock was nearly identical at this dataset size and on a single node (8.73 s broadcast vs. 8.85 s sort-merge), but the physical plans differ fundamentally: broadcasting the 18-row `airlines_dim` table avoids re-partitioning the 3M-row flights table entirely, a saving that grows with both cluster size and network cost.

5. **Adaptive Query Execution (AQE) provides automatic runtime tuning.** With `spark.sql.adaptive.enabled = true`, Spark dynamically coalesces post-shuffle partitions, avoiding the overhead of processing

many empty partitions when data is skewed. Our scalability sweep (Section 4.4) shows Q3 plateaus at ~ 0.5 s once partitions exceed core count – AQE keeps that plateau stable instead of letting the curve degrade with too-large defaults.

6. **Caching pays for itself after one re-execution.** Section 4.1 measured `flights.cache()` delivering a 13.5× speedup on Q2 (5.08 s cold vs. 0.38 s warm). For interactive analytics where the same dimension table is hit by many queries, caching is the single highest-ROI optimization in Spark – it eliminates CSV decoding entirely for every subsequent query.

7. **File format matters even without filters.** Section 4.2 showed Parquet ran 5.71× faster than CSV on Q3 (1.02 s vs. 5.82 s) despite the query having no predicate to push down. The columnar layout still lets Spark read only the three columns the aggregation touches and skips per-row text-to-double parsing. Combined with partition pruning, filtering on `YEAR=2022` reads only 687,860 of the 3M rows (4.4× less data) – a saving that scales linearly with cluster size and dataset growth.

8. **RDDs are best reserved for unstructured data and custom transformations.** For all structured analytical queries in this benchmark, RDDs were uniformly slower, more verbose, and lacked automatic optimization. Their role in modern Spark applications should be limited to scenarios where the Structured API cannot express the required transformation.

8 Code Submission Layout

All terminal-runnable code lives in the `scripts/` directory alongside this report. Each query is a standalone `spark-submit` target and produces its own log under `scripts/logs/`; the captured CSV in `scripts/results/` is the exact source of the performance table in Section 6.

```
mini Project 2/
|-- main.ipynb               # original Colab notebook (reference; not required to run)
|-- flights_sample_3m.csv    # dataset (3M rows, ~600 MB)
|-- Report/
|   |-- main.tex             # this report
|   `-- *.png                # query + explain screenshots
`-- scripts/
    |-- README.md            # WSL setup and run instructions
    |-- common.py            # SparkSession factory, schema, helpers, perf logger
    |-- download_dataset.py  # one-time Kaggle pull
    |-- run_all.sh           # spark-submit each script + aggregate results
    |-- aggregate_results.py # build performance_pivot.csv from the raw log
    |-- q01_filter.py ... q12_anomaly.py    # 12 query scripts (RDD+DF+SQL+explain)
    |-- optimization/
    |   |-- caching.py             # cold vs warm Q2
    |   |-- parquet_format.py      # CSV vs Parquet on Q3
    |   |-- partition_pruning.py   # YEAR=2022 partition prune on Parquet
```

```
|   `-- scalability.py         # shuffle.partitions sweep
|-- logs/                 # tee'd stdout per script (incl. .explain blocks)
`-- results/              # performance_results.csv, performance_pivot.csv
```

How to reproduce

```
# WSL2 / Ubuntu 22.04 prerequisites
sudo apt install -y openjdk-17-jre-headless python3 python3-pip python3-venv

cd "mini Project 2"
python3 -m venv .venv && source .venv/bin/activate
pip install pyspark==3.5.1 kagglehub          # pandas optional

cd scripts
./run_all.sh                                  # runs all 12 queries + 4 optimization studies
python3 aggregate_results.py                  # prints RDD vs DF vs SQL pivot
```

Per-query reproduction (used for the screenshots in Section 3) is a single command:

```
spark-submit --master "local[4]" --driver-memory 4g \
             --py-files common.py q09_broadcast_join.py
```

Cluster configuration knobs

The runner honours the following environment variables for scalability tests:

Variable	Default	Effect
SPARK_MASTER	local[4]	cluster URL / cores
DRIVER_MEMORY	4g	driver heap
SHUFFLE_PARTITIONS	200	spark.sql.shuffle.partitions
FLIGHTS_CSV	project root	source CSV path
FLIGHTS_PARQUET	project root	Parquet output / read path
RESULTS_DIR	scripts/results	where the CSV log is written

For example, a 2-core run with 100 shuffle partitions:

```
SPARK_MASTER='local[2]' SHUFFLE_PARTITIONS=100 ./run_all.sh
```